



$$a_n = \frac{(-5)^n}{(n+3)8^{n+1}}$$

1) Find $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$.

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{5}{8}$$

2) Based on your answer, what does the ratio test say about $S = \sum_{n=1}^{\infty} a_n$?

Choose 1 answer:

☒ CORRECT (SELECTED)
The series converges.

☐ INCORRECT
The series diverges.

☐ INCORRECT
The test is inconclusive.

1 / 2

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{5}{8}$$

Show entire process. ✓

2 / 2

Since the limit, $\frac{5}{8}$, is less than 1, the ratio test tells us that the series converges.

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Ratio test

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